

How to start Møller Polarimeter DAQ with CODA 2.2 on Linux

Aug 04 2006, v2.3

This text was adapted for Møller Polarimeter DAQ from [Hints on Hall A Data Taking with CODA 2.2 from Bob Michaels](http://www.jlab.org/~moller/guide1.2_linux.html). This file: www.jlab.org/~moller/guide1.2_linux.html

I. How to Run CODA 2.2 Møller Polarimeter DAQ on Linux box

Preface

To run CODA on linux box we are using ‘adaql1’ computer and local home directory for moller account: /adaqfs/home/moller !!!

The MSQL database server runs on computer ‘adaql1’. The name of database is ‘mollerpol_1’ and session name is ‘moller’. The MSQL database listening port (MSQL_TCP_PORT) set to 8301. The database is stored in directory /adaqfs/home/moller/MSQL (since June 2004). If there is no MSQL server running on adaql1 one should login to adaql1 using ‘moller’ account and type to start it: `run_daemon_db msqld &`

We are using different database for running DAQ on Sun’s and Linux mashines. To change database one should modify boot file for Moller Detector VME crate (hallavme5): `~moller/vxworks/hallavme5_c2.boot`. Uncomment (delete symbol ‘#’ at start of line) next lines:

- `#putenv "EXPID=mollerpol_1"` — set name of MSQL database
- `#putenv "CMLLOG_PORT=8229"`
- `#putenv "CMLLOG_HOST=adaql1"`
- `#bigendian_out=0;` — set bytes order to little endian(PC)

Reboot VME crate ‘hallavme5’.

Then need to synchronize run number for this database. To do this one should start ‘dbedit’, select database with name “mollerpol_1” select table “sessions” and modify run number to the last run number in database “mollerpol”.

To change bytes order there is another possibility to do this remotely: Type on any of ‘adaq**’ computer ‘dpwish’. Then in dpwish terminal type next command: `DP_ask ROC11 configure -bigendian_out 0` To check value of ‘bigendian_out’ type next command: `DP_ask ROC11 cget -bigendian_out`

ROC11 - is name of ‘hallavme5’ in MSQL database. Value 1 is corresponded ROC connections to big endian mashine(Sun) Value 0 is corresponded ROC connections to little endian mashine(PC).

First start CODA on ‘adaql1’ computer.

1. To start CODA on the workstation, type “start_coda”.
2. Windows with ‘et’, ‘coda_er’, ‘coda_eb’, ‘rcServer’, ‘runcontrol’, will popped-up. Also ‘cdevGateway’ server for slow control will started up.
3. Next start slow control.
4. Use the “moller” account on “adaql1” or “adaql2” computer.
5. [Start Møller Polarimeter Slow Control program](#). On “adaql1” change directory to ‘~/Java/msetting’ and type “mpc”. The settings parameters for detector electronics loaded from file: /adaqfs/home/moller/daq/coda2/DetectorSettings/DETSET_new

Note: There is also ‘mollerSetting’ program for detector electronics setting on Linux platform. If ‘mollerSetting’ is used instead of the program for detector electronics setting in ‘mpc’ the last will not working and one need to modify script: ‘/adaqfs/home/moller/daq/coda2/scripts/getruninfo’ The next line should be uncomment (remove symbol ‘#’ at begin of line):

```
# if [ 0 -gt 1 ]
```

And the next line should be comment (place symbol ‘#’ at begin of line):

```
if $jv -classpath /adaqfs/home/moller/Java/msetting MPCclient $h $p "get  
detectorinfo" > $CODAPATH/RunInfo/detector.set
```

It is necessary to get information about detector settings from right source.

6. When CODA ‘runcontrol’ window pops-up, press the Connect button. After “connect”, wait 20 seconds, then to select or change configurations, use the “Configuration” button. Choose the configuration from the ‘Run Type’ dialog box, then press “Download”. The relevant configurations are:
 - **beam_pol** — to make beam polarization measurements, (the sign of current in Helmholtz coils change to opposite for every run)
 - **moller_data** — the same as configuration ‘beam_pol’, but current in Helmholtz coils setup to zero.
 - **moller_puls** — to do a pulser run, testing of DAQ, and get pedestals.
 - **moller_LED** — to do a run with flashing LED, and detector testing.If you’ve been running, you will first have to push “Abort” button.
7. To start and stop runs, push the buttons “Start Run” and “End Run”. Start Run is the same as the sequence prestart, go. After starting run you getting data with events rate about 300-500 Hz, and data rate 30-50 kByte/sec.

Where is the data?

The all data save to the directory /adaql1/data1/moller/

The names of data files for every type of run are:

Run type	file name
beam_pol	moller_data_NN.dat
moller_data	moller_data_NN.dat
molelr_puls	moller_data_puls_NN.dat
molelr_LED	moller_data_led_NN.dat

Where NN is number of run.

Disk Handling?

Storage to MSS and purging of disks is done automatically. Things to *NOT* do:

- Do *NOT* erase or move data files (even if you think its junk).
- Do *NOT* try to change run numbers.
- Do *NOT* change names of run files.
- Do *NOT* try to put data by hand into MSS.

It takes a couple hours for files to appear in MSS, and the files remain on adaq disk for typically 1 or 2 days.

To end working with CODA press “DISCONNECT” button on ‘runcontrol’ window and type ‘kcoda’ in the terminal window.

II. Resets of CODA and Cold Starts

Problems with CODA 2.2 can usually be solved with a simple reset or with a Cold Start. If not call Bob Michaels or Dave Abbott.

RESETS

The DAQ uses hallavme5(ROC11). The halladaq14 is used only for target control (to set current in Helmholtz coils, to do the target magnetization measurement and the temperature of target measurement). If the ROC seems really frozen, press the green button on the panel of buttons in the center room of the counting house. The buttons are labelled “moller detector VME #2”, “moller target VME#1”. Hold the button about 10 seconds. If you reboot the ROC, or if something on the workstation is hung up, try pushing the “Reset” button in runcontrol (NOTE: If “cancel” button is available, you should first push “cancel” before trying “reset”, else reset may do nothing). Then configure by pressing “Configure” and choosing the configuration you want (presumably the “beam_pol”), download, and Start Run. That should work, but if not try a cold start (see below).

COLD START

First kill all process on the workstation (adaql1) by typing from anywhere on the adev account “kcoda”. This stops runcontrol, the event builder, event recorder, the runcontrol server, and cleans up the ET system. Now you can start everything again by typing “runcontrol” but it is first a good idea to make sure the VME crates (i.e. the ROCs) are running. One finds that when resetting, one can frequently avoid rebooting the frontend crates, and just restart runcontrol, but if you must: Press the green buttons for ROC11 as explained in the Reset section above. Then press the Connect button. After “connect”, wait 20 seconds, then press “Configuration” and then “Run Type”; a dialog box pops up and you must choose the configuration you want, which is presumably “beam_pol” for make beam polarization measurement. Then press download and wait about 30 seconds. Now you can “Start Run” to start a run. On rare occasions you may need to “Reset” after configuring the “Run Type” (if mysql database gets confused about state of things).

THINGS THAT GO WRONG WITH DAQ or COMPUTERS

1. Lots of deadtime — As of July 98, we found a bad cable in the SCSI chain which tricked the operating system into lowering the rate on SCSI. With the solved, we hopefully won’t see deadtime fluctuations at high rate when we write to MSS. Still, there are a few other causes: a) Rapid disk I/O to the same disk partition as where we write data. We avoid this now. b) Huge event sizes. Normal is ~1 kbyte. If much larger, then maybe you have no pedestal suppression, or oscillations in preamps, pileup in TDC, etc.
2. Computer problems: First try calling Bob Michaels. You may also call the computer-center expert on-call (ask the guard shacalling Bob Michaels. You may also call the computer-center expert on-call (ask the guard shack, extension 4444, to do it). If Bob Michaels isn’t around, try calling Mark Jones or Dave Abbott for CODA problems.

III. How to Program the RAM of the ROCs

This is a skill you RARELY need but I add it here for completeness. You may skip reading the rest of this.

Since we rotated VME computers in July 98 (the newest fastest are deployed, the oldest slowest are spares), and since the 100 Mb VME computers are on their own subnet, the old CODA 1.4 setup is dead. It could be revived, if needed. Here's the 2.2 setup

To connect to VME crates the serial port server is used. Type `telnet hatsv5 2004` for connection to 'hallavme5'. Type `telnet hatsv5 2003` for connection to 'halladaq14'. Enter username: `adaq` And password: `*****`

For hallavme5

This is for Moller detector DAQ

hallavme5 VxWorks boot

VxWorks (for Motorola MVME2300 - MPC 604r) version 5.3.1. Kernel: WIND version 2.5.	
boot device	dc
host	adaql2
file	adaql2:/adaqfs/home/moller/vxworks/vx2306_1
(e)	129.57.192.31:ffffff00
(h)	129.57.164.59
(g)	129.57.192.1
(u)	moller
(f)	0x20
(tn)	hallavme5
(s)	/adaqfs/home/moller/vxworks/hallavme5_c2.boot

For halladaq14

This is for Moller target control

halladaq14 VxWorks boot

VxWorks (for Motorola MVME162) version 5.2. Kernel: WIND version 2.4. Made on Thu Aug 1 12:46:31 EDT 1996.	
boot device	ei(0,0)
host	adaql2
file	adaql2:/adaqfs/home/moller/vxworks/vx162-8MB
(e)	129.57.164.39
(h)	129.57.164.59
(u)	moller
(f)	0x20
(tn)	halladaq14
(s)	/adaqfs/home/moller/vxworks/halladaq14.boot

Reminder of vxWorks operations.

You are either in the mode with a prompt like `[VxWorks Boot]:` or a prompt `->`

In the former case, you change the NVRAM by typing “c” (without my quotes), and typing the new fields to the right of the old fields. To reboot, you type “@”.

In the latter case you type at the -> prompt, “bootChange” to change NVRAM, and type “reboot” to reboot.